Lecture 2

Started: Sep 25 at 5:25pm

Quiz Instructions

Question 1 — 1 pts

What is in the problem space of the traveling salesman problem?

- (●) The visited cities.
- ( ) The decisions of the AI salesman.
- ( ) The bought products.
- ( ) The sold products.

Question 2 — 1 pts

What does the state space of the Hanoi tower problem contain?

- ( ) Sequences of moves to move the disks.
- ( ) The number of the disks on each peg.
- ( ) The identifiers of the disks on each peg, ordered by the pegs.
- (●) The identifiers of the pegs each disk is on, ordered by the disks.

Question 3 — 1 pts

What does the problem space of the Hanoi tower problem contain?

- ( ) The number of the disks on each peg.
- ( ) The sequence of moves needed to move the disks from the third peg to the first peg.
- (●) Sequences of moves to move the disks.
- ( ) The identifiers of the disks on each peg, ordered by the pegs.

Question 4

1 pts

What does the solution in the problem space of the Hanoi tower problem contain?

- (●) The sequence of moves needed to move the disks from the third peg to the first peg.
- ( ) The number of the disks on each peg.
- ( ) The identifiers of the disks on each peg, ordered by the pegs.
- ( ) The identifiers of the pegs each disk is on, ordered by the disks.

Question 5

1 pts

In how many seconds did Bobby Fischer solve the 15-puzzle (a more difficult variant of the 8-puzzle) on the Tonight Show? You may want to watch this video: https://www.youtube.com/watch?v=QxvnEwvgfeI

- ( ) 187
- ( ) 32
- ( ) 126
- (●) 17

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Lecture 3

Started: Oct 2 at 11:28pm

Quiz Instructions

Question 1 — 1 pts

Is the hill climbing method going uphill or downhill?

- ○ It depends on the second derivative of the objective function.
- ○ Uphill.
- ○ It depends on the gradient of the objective function.
- ◉ Downhill.

Question 2 — 1 pts

Can the hill climbing search go on forever without finding the goal?

- ○ No, because it would need an equidistant surface, which is impossible.
- ◉ Yes, if neighboring nodes have identical values.
- ○ No, because it constantly monitors already visited nodes and detects cycles.
- ○ Yes, if it first goes downhill, then uphill.

Question 3 — 1 pts

What is the tabu set in tabu search?

- ○ A list of nodes which shouldn't be expanded.
- ○ A list of nodes which are completely excluded from the search.
- ○ A list of nodes which shouldn't be visited.
- ◉ A list of the last few current nodes.

Question 4 1 pts

What real-world process does simulated annealing simulate?

- (●) The forging of swords.
- ( ) The preparation of steaks in fine dining.
- ( ) The heating of materials.
- ( ) The cooling of liquids.

Question 5 1 pts

What is the most important component for local search algorithms that guides them to the goal?

- (●) The heuristics.
- ( ) The goal table.
- ( ) The solution matrix.
- ( ) The iterations.

Quiz saved at 11:28pm Submit Quiz

Lecture 4

Started: Oct 2 at 11:32pm

Quiz Instructions

Question 1 — 1 pts

Where does BT1 keep track of the path from the start node to the current node?

- ○ In a hidden global stack.
- ◉ Implicitly, via the parameters of the recursive function calls.
- ○ In a private local stack.
- ○ Implicitly, via the solution variable in each recursive function call.

Question 2 — 1 pts

How does BT1 know whether an outgoing edge from a node was already traversed or not?

- ○ Based on A-star.
- ○ Based on whether the node is part of the solution variable or not.
- ○ Based on the current parameters of the recursive function calls.
- ◉ Based on which nodes were already iterated over by the for loops in each recursive function call.

Question 3 — 1 pts

What is main the difference between the 2nd and 3rd state-space model for n-queens when backtracking?

- ○ In the 2nd, we can place queens anywhere. In the 3rd, we place the queens row-by-row.
- ○ In the 2nd, we place all queens at once and then change their position until we find the solution. In the 3rd, we place the queens one-by-one.

- (●) In the 2nd, we don't check whether a queen that we place attacks other queens. In the 3rd, we can only place new queens so they don't attack existing ones.
- ( ) In the 2nd, we can place queens anywhere. In the 3rd, we check that there should be no existing queen where we want to place the new one.

Question 4 — 1 pts

Which of these is not a heuristics for solving n-queens by backtracking?

- (●) Row-by-row
- ( ) Number of attacked squares
- ( ) Odd-even
- ( ) Diagonal

Question 5 — 1 pts

What does the AC1 algorithm do?

- (●) It runs Look Forward until there was no elimination.
- ( ) It runs Look Forward until it finds the solution.
- ( ) It runs Forward Checking until there was no elimination.
- ( ) It runs Forward Checking until it finds the solution.

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Lecture 5

Started: Oct 16 at 5:01pm

Quiz Instructions

Question 1 — 1 pts

Which of these is not a function used in graph search?

- ○ the evaluation function
- ○ the parent pointer function
- ◉ the open node function
- ○ the cost function

Question 2 — 1 pts

When do we put a node into the set of open nodes?

- ◉ At the start of the search, when we encounter a node we've not seen before, and when we find a cheaper path to an already seen node.
- ○ At the start of the search and when we encounter a node we've not seen before
- ○ When we encounter a node that's not already in the set of open nodes and has optimal cost.
- ○ When we encounter a node which takes us closer to the goal node.

Question 3 — 1 pts

Which of these graph search algorithms find the optimal (cheapest) solution?

- ○ depth-first search
- ○ A algorithm
- ○ breadth-first search

- ◉ uniform-cost search

Question 4 — 1 pts

Which of these graph search algorithms uses a heuristic?

- ○ breadth-first search
- ◉ A-star algorithm
- ○ uniform-cost search
- ○ depth-first search

Question 5 — 1 pts

Why do we use heuristics in graph search?

- ◉ They guide our search towards the goal, so we can find it faster.
- ○ They guide our search towards the goal, so we can find it more reliably.
- ○ They allow us to incorporate expert knowledge about the problem into the search model so we can find a better (cheaper) solution.
- ○ They guide our search towards the goal, so we can find a better (cheaper) path to it.

Quiz saved at 5:01pm Submit Quiz

Lecture 6

Started: Oct 16 at 4:59pm

Quiz Instructions

Question 1 — 1 pts

What is a zero-sum game?

- ○ The sum of the rewards the players get converges to zero.
- ○ The sum of the costs along the state-graphs of the players is zero.
- ○ The sum of the expected rewards the players get converges to zero.
- ◉ The gain of one player is the loss of the other player, so the sum of their payoffs at the end of the game is zero.

Question 2 — 1 pts

Where do the games end in a game tree?

- ○ At the branches.
- ○ At the final level.
- ○ At the root.
- ◉ At the leaves.

Question 3 — 1 pts

What is a winnig strategy in a game tree?

- ○ A set of leaves where the player wins.
- ○ A set of decisions of what to do at each node.
- ○ A winning play of the game.

◉ A beam of plays leading to win.

Question 4 1 pts

Why do we need an evaluation function?

- ○ Because otherwise we couldn't obtain a winning strategy.
- ◉ Because building the whole game tree is computationally not feasible.
- ○ To evaluate all of the nodes of the game tree at once.
- ○ Because we can evaluate the utility of the nodes more precisely with it.

Question 5 1 pts

Which of these is the most useful and widely used variant of the minimax algorithm?

- ○ (m,n)-average evaluation.
- ◉ Alpha-beta algorithm.
- ○ Negamax algorithm.
- ○ Various depth bound evaluation.

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Lecture 7

Started: Oct 22 at 9:10pm

Quiz Instructions

Question 1 — 1 pts

Which of these is not an evolutionary operator?

- ○ recombination
- ◉ fitness
- ○ selection
- ○ mutation

Question 2 — 1 pts

What does the fitness function measure?

- ◉ How close an individual is to the solution.
- ○ How close a population is to the solution.
- ○ The robustness of an individual with respect to evolutionary changes.
- ○ How fit the population is with respect to evolutionary changes.

Question 3 — 1 pts

Which of these problems was not phrased as an evolutionary problem in the lecture?

- ◉ 8-puzzle
- ○ traveling salesman
- ○ rock-paper-scissors
- ○ n-queens

Question 4 — 1 pts

What is partial crossover trying to solve?

- ○ The problem of partial information when we cannot see the whole representation.
- ◉ The duplication of signals when we have permutations as representations.
- ○ The problem when computing the whole crossover is not feasible.
- ○ Sometimes we cannot do a whole crossover, only a partial one.

Question 5 — 1 pts

Which of these is not a valid selection method?

- ◉ card shuffle
- ○ culling
- ○ roulette wheel
- ○ tournament

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Lecture 8

Started: Nov 2 at 3:08pm

Quiz Instructions

Question 1 — 1 pts

Which of these is not a classification problem?

- ○ Determining the digits of a postal code on a handwritten envelope.
- ○ Determining the grade for a programming assignment.
- ◉ Determining the asking price of a house we would like to sell.
- ○ Determining the category of a news article (sports, politics, etc.).

Question 2 — 1 pts

Why do we need test examples for a neural network?

- ○ To test for bugs in the implementation of the network.
- ○ To test how well the network performs on data across different user groups.
- ○ To test the training performance of different network architectures.
- ◉ To test how well the network performs on unseen data, which helps us estimate how well it will perform in the real world.

Question 3 — 1 pts

Which of these is not a component of a neural network?

- ○ loss function
- ○ activation function
- ◉ input-output pairs

- ○ weight matrix

Question 4 — 1 pts

Which of these is the fastest for training a typical large neural network?

- ◉ TPU
- ○ FPU
- ○ CPU
- ○ GPU

Question 5 — 1 pts

Which of these is not a Python library used in machine learning workflows?

- ○ scipy
- ◉ grumpy
- ○ matplotlib
- ○ numpy

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Lecture 9

Started: Nov 14 at 5:11pm

Quiz Instructions

Question 1 — 1 pts

Which python package handles dataframes?

- ○ numpy
- ◉ pandas
- ○ sklearn
- ○ nltk

Question 2 — 1 pts

Which of these is not a step in creating and training a neural network?

- ◉ cleaning the model
- ○ defining the model
- ○ compiling the model
- ○ fitting the model

Question 3 — 1 pts

Why is overfitting bad?

- ○ Because we will have too low error on the training data so when new data comes we cannot train any further.
- ○ Because our performance is good in the real world, but bad on the benchmarks that matter.
- ○ Because our performance will be bad in the real world even though it's good on the test data.

- [x] Because our performance will be bad in the real world even though it's good on the training data.

Question 4 — 1 pts

Why are tensors important in deep learning?

- [x] Because in most neural network architectures, the weights are stored in tensors.
- [ ] Because the name of the most well-established deep learning framework by Google is TensorFlow.
- [ ] Because of the deep interaction of deep learning and tensors from physics.
- [ ] Becaue the best hardware for deep learning is called TPU (Tensor Processing Unit).

Question 5 — 1 pts

How do we compute backpropagation for the neural networks?

- [ ] We usually have to derive the formulas on paper and then code them. We need to take the derivative of the loss function with respect to the weights.
- [ ] We usually have to derive the formulas on paper and then code them. We need to take the derivative of the network with respect to the weights.
- [ ] We usually have to derive the formulas on paper and then code them. We need to take the derivative of the loss function with respect to the weights and biases.
- [x] We don't have to, TensorFlow already does that automatically.

Quiz saved at 5:11pm Submit Quiz

Lecture 10

Started: Nov 14 at 5:12pm

Quiz Instructions

Question 1 — 1 pts

Why do we do exploratory data analysis and data cleaning?

- ○ Because we would like to get an understanding of the data.
- ○ Because if we give garbage data to the model it will give us garbage results.
- ◉ All of the answers are true.
- ○ Because it can help us choose the right model.

Question 2 — 1 pts

What is a stopword?

- ◉ A word we don't consider relevant to the problem.
- ○ A word where we stop processing the input data.
- ○ A word which denotes the boundaries between units of text.
- ○ A word where the model needs to stop and make a decision.

Question 3 — 1 pts

Which of these is not a commonly used language model?

- ○ n-gram
- ○ bigram
- ○ trigram
- ◉ pentagram

Question 4 1 pts

What kind of hyperplane does the SVM obtain?

- ○ One whose distance to the closest positive and negative examples are minimal.
- ○ A maximal merging hyperplane.
- ○ A Vapnik-hyperplane.
- ◉ One whose distance to the closest positive and negative examples are maximal.

Question 5 1 pts

Which is not an important component of most machine learning workflows?

- ◉ stemming
- ○ training
- ○ hyperparameter search
- ○ cross-validation

Quiz saved at 5:14pm Submit Quiz

Lecture 11

Started: Nov 21 at 2:05am

Quiz Instructions

Question 1 — 1 pts

Which is a common combination for multiclass classification?

- ○ sigmoid and binary cross-entropy
- ○ softmax and binary cross-entropy
- ○ sigmoid and categorical cross-entropy
- ◉ softmax and categorical cross-entropy

Question 2 — 1 pts

Which of these is not an NLP problem?

- ○ POS tagging
- ○ named entity recognition
- ○ tokenization
- ◉ LALR parsing

Question 3 — 1 pts

What are word embeddings used for?

- ○ to determine how similar the meaning of two words are
- ○ to make dense distributed vector representations useable by neural networks
- ◉ all of the answers are true
- ○ to utilize the computational power that was used to train downloadable word embeddings

Question 4 1 pts

What is the shape of the word embedding matrix we used?

- ○ (number of documents in the corpus, number of words in the vocabulary)
- ○ (300, number of words in the vocabulary)
- ◉ (number of words in the vocabulary, 300)
- ○ (number of words in the vocabulary, number of documents in the corpus)

Question 5 1 pts

What is a Keras?

- ○ a female death-spirit
- ○ a person's fate
- ○ a deep learning framework
- ◉ all of the answers are true

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Lecture 12

Started: Nov 28 at 12:56am

Quiz Instructions

Question 1 — 1 pts

Whose works did we try to replicate with an RNN?

- ○ George R.R. Martin
- ○ Guy Gavriel Kay
- ○ J.R.R. Tolkien
- ◉ William Shakespeare

Question 2 — 1 pts

What kind of RNN did we use?

- ○ LSTM
- ○ vanilla RNN
- ◉ GRU
- ○ GNU

Question 3 — 1 pts

What did we predict based on previous text?

- ◉ the next character
- ○ the next sentence
- ○ the next token
- ○ the next word

Question 4 1 pts

What kind of layer did we not have in the model?

- ○ GRU
- ○ Dense
- ○ Embedding
- ◉ Conv2D

Question 5 1 pts

What are checkpoints good for?

- ◉ to save the weights of the model periodically
- ○ to check the correctness of the data periodically to ensure proper training
- ○ to check our progress peroidically by monitoring validation accuracy
- ○ to check our progress peroidically by monitoring training accuracy

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Lecture 13

Started: Dec 5 at 12:12am

Quiz Instructions

Question 1 — 1 pts

What is the main difference between unsupervised and supervised learning?

- (●) for unsupervised learning we don't have labels in the training data
- ( ) for unsupervised learning we don't have data points in the training data
- ( ) for supervised learning we have a machine (AI) supervisor who helps the algorithm
- ( ) for supervised learning we have a human supervisor who helps the algorithm

Question 2 — 1 pts

Which of these is an unsupervised learning problem?

- ( ) determining the topic of a document from a few predefined categories
- (●) determining the most prevalent topics in some documents
- ( ) determining the selling price of a house
- ( ) determining the digits on handwritten envelopes

Question 3 — 1 pts

What kind of algoritm is k-means?

- ( ) soft clustering
- ( ) hierarchical clustering
- (●) hard clustering
- ( ) overlapping clustering

Question 4 — 1 pts

Where is Singular Value Decomposition (SVD) useful?

- ○ for recommender systems
- ○ for topic modeling
- ○ for PCA
- ◉ all the answers are true

Question 5 — 1 pts

Why should there be a bottleneck in an autoencoder?

- ◉ so it learns some useful representation
- ○ to make the training separable to multiple GPUs
- ○ to reduce the possibility of overstepping
- ○ to make the training more even (smooth)

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